WORLD MODELS / EMBODIED AI

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World models, embodied AI, and active learning, with a focus on data efficiency and physical consistency.

01 Education

- **Tsinghua University •Qiuzhen College** 2025 - Present
  M.Sc. student, Mathematics & Artificial Intelligence. Advisor: Changshui Zhang; close collaborator: Sen Cui.
- **Southeast University •School of Mathematics** 2021 - 2025
  B.Sc. in Mathematics & Applied Mathematics. Honors include SEU Undergraduate Innovation Star, National First Prize in the Chinese Mathematics Competitions, and First Prize in the Jiangsu Division of CUMCM.

02 Competitions & Awards

- **Tsinghua University Graduate Comprehensive Excellence Scholarship** **First Class** Oct 2026
- **Zhongguancun Embodied AI Robotics Competition •Outstanding Participating Team** Nov 2025
  2nd edition, BAAI Embodied Brain Large Model Challenge; data curation and RoboBrain2.0 fine-tuning/evaluation.

03 Work & Internship Experience

- **Beijing Academy of Artificial Intelligence (BAAI)** Zhiyuan Scholar Intern ·World Models & Embodied AI Mar 2026 - Present
  Research with Sen Cui on embodied data generation, confidence-guided active learning, and counterfactual motion reasoning. Work on datasets, methods, and experiments to improve data efficiency, physical consistency, and long-horizon prediction.

04 Projects & Research

Lead Research

- **ConfAL-WM: Confidence-Guided Active Learning for Action-Conditioned World Models**
  Xiang Liu[*], Sen Cui[†], Changshui Zhang[†], *et al.* | NeurIPS 2026 ·3 Workshops | arXiv:2608.25572
  Developed a confidence probe that guides task selection and local training using frame- and patch-level scores. Improved RoboTwin2.0 post-training at a 40% data budget, with gains in reconstruction and semantic quality.
- **LPA-CWM: A Learned Physical Adjudicator for Motion Reasoning with Counterfactual World Models**
  Kunwei Wu[*], **Xiang Liu**[*], Sen Cui[†], *et al.* | arXiv:2609.14073
  Developed a 3.0M-parameter adjudicator to weight counterfactual motion responses by reliability. Established CMC evaluation, improving average Dynamic Correspondence Accuracy by 18.1%-60.0% over uniform aggregation on DAVIS, Kinetics, and RoboTAP.
- **Scene2Demo: Self-Evolving Embodied Data Generation via Object-Action Graph**
  Xiang Liu[*], Sen Cui[*†], Guocai Yao[†], Changshui Zhang[†], *et al.* | FMEA @ CVPR 2026 ·Poster | arXiv:2602.12065
  Built an image-and-query-to-demonstration pipeline using object-action graphs and visual feedback. Achieved 71.6% execution success across 102 primitive scene-task pairs, generating simulation data for downstream policy learning.

Collaborative Research

- **HamiFormer: Dual-Expert Diffusion Fields with Affine Symplectic Maps**
  Haoxiang Huang[*], **Xiang Liu**, Sen Cui[†], *et al.* | arXiv:2609.32838
  Combines diffusion and symplectic Hamiltonian propagation with adaptive routing for stable long-horizon prediction.
- **GeoCo-SAVi: Geometry-Consistent Slot Attention for Explicitly Editable Object Representations**
  Haoxiang Huang[*], Zhekai Wang[*], **Xiang Liu**, Sen Cui[†], Changshui Zhang[†], *et al.* | arXiv:2609.06628
  Uses equivariant decoding and appearance transplantation for editable object position and scale with geometric consistency.
- **MOSH-WM: Mask-Grounded Soft-Hamiltonian Dynamics for Object-Centric World Models**
  Zhekai Wang[*], Haoxiang Huang[*], **Xiang Liu**, Sen Cui[†], *et al.* | arXiv:2608.22750
  Uses mask-grounded object states and soft-Hamiltonian priors to improve long-horizon closed-loop prediction.